Data Science

Lecture 07: Data Exploration and Preprocessing

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Speaker notes

Converted from LNKB-tex part-en-ds-chp3-data.tex (two source units re-encoded from ISO-8859). Uses Lecture 02's quantiles and covariance throughout, and its random variables and joint distribution in the unit on data as random variables.

Where We Left Off

Three lectures fitted models to tables nobody questioned. Today steps back to KDD steps 1–2 from Lecture 01: what the columns are, whether they can be trusted, and what a plot shows that a summary hides.

Lecture 01
data is never raw

→

Lecture 02
summaries can agree while the data differs (Anscombe)

→

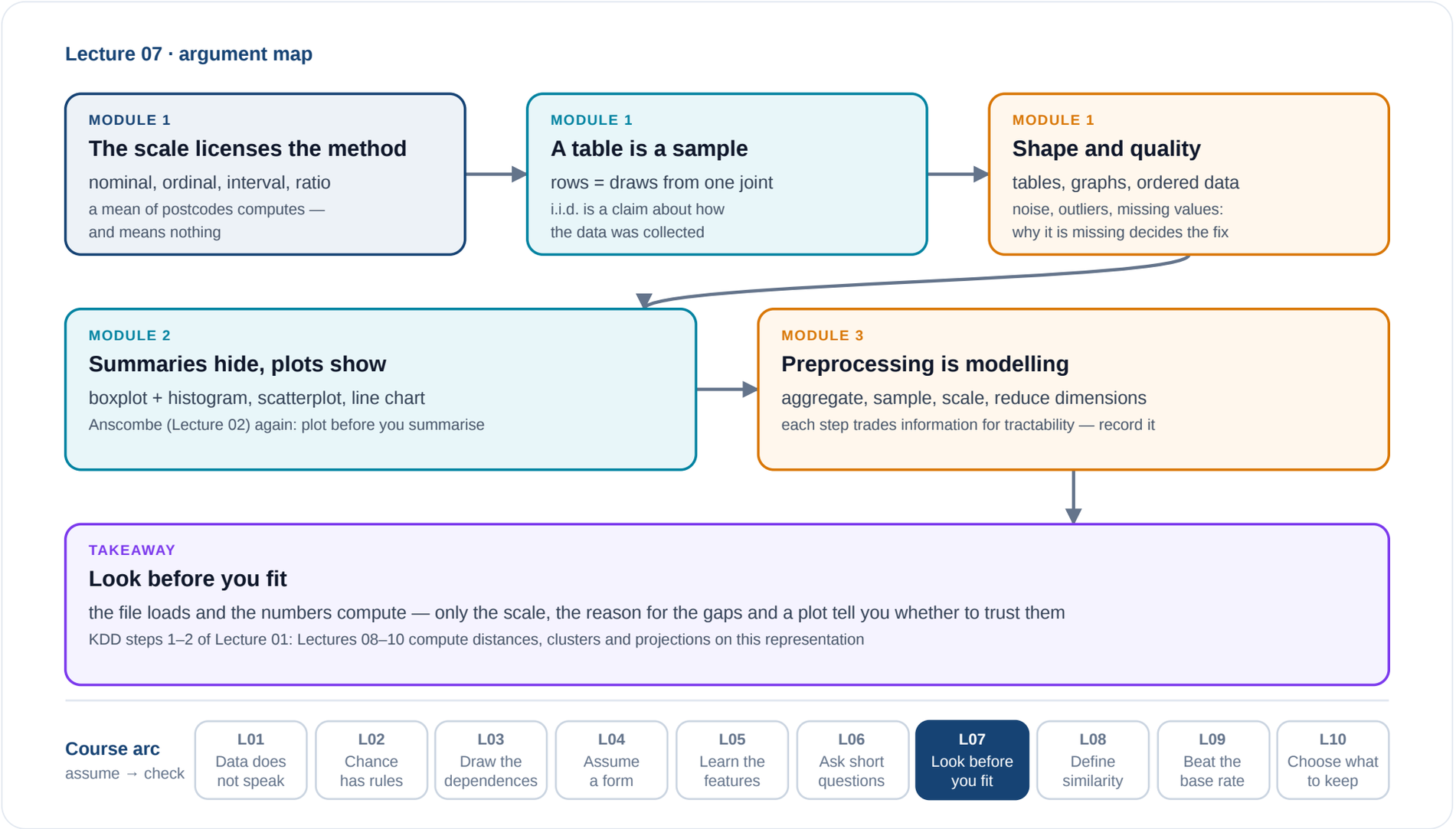
Today
scale, quality, plots, preprocessing

> From Lecture 08 on there is no label to check a model against — looking first stops being good practice and becomes the main check we have.

Speaker notes

Name the order openly. Modelling came first because judging data needs the language (Lecture 02) and the models (Lectures 04–06); now the course runs the loop backwards, as the second roadmap slide of Lecture 01 showed. Attribute types were used informally in Lectures 04–06 (nominal attributes in naive Bayes and trees); today they get their names and their consequences.

Argument Map



Speaker notes

The scale of measurement licenses a method: a mean of postcodes computes and means nothing. A table is a sample from a joint distribution, and "i.i.d." is a claim about how it was collected. Why values are missing decides how they may be filled. Plots show what summaries hide. Preprocessing is modelling: every aggregation, sample, scaling and reduction trades information for tractability. The takeaway is the lecture's arc chip.

Module 1

What Kind of Data Is This?

> ***Guiding Question:*** *The file loads and the numbers compute. Why is that not yet a reason to trust the average?*

Four Attribute Types — What the Values Allow

	type	the values…	meaningful	statistics	examples
categorical	**nominal**	are names that only distinguish objects	$=, \neq$	mode, entropy, contingency table, χ^2 test	ZIP code, employee ID, eye colour
	ordinal	carry an order	$<, >$	median, percentiles, rank correlation, sign test	grades, rankings, {good, better, best}
numeric	**interval**	have meaningful differences; the zero is arbitrary	$+, -$	mean, standard deviation, Pearson correlation, t-test	temperature in °C or °F, calendar dates
	ratio	have a true zero, so ratios are meaningful	$\times, \div$	geometric mean, harmonic mean, coefficient of variation	length, mass, counts, age, temperature in K

Each type permits everything the one above it does, plus more.

Speaker notes

The four scales go back to S. S. Stevens (1946). Nominal and ordinal attributes are called categorical (qualitative), interval and ratio attributes numeric (quantitative). The scales form a hierarchy: each permits the comparisons and statistics of the scales above it in the table, plus its own. So the mode is meaningful on every scale, the median from ordinal on, the mean from interval on, and ratios, the geometric mean and the coefficient of variation only on a ratio scale.

The type is a property of the measurement, not of the storage format: a ZIP code stored as an integer is still nominal, and a street number is ordinal at best. Temperature in kelvin, unlike °C, is a ratio scale.

Four Attribute Types — What You May Transform

	type	permissible transformation	example
categorical	**nominal**	any one-to-one relabelling	reassigning employee IDs changes nothing
	ordinal	any strictly increasing f: $x \mapsto f(x)$	{good, better, best} is as well coded by {1, 2, 3} as by {1, 5, 9}
numeric	**interval**	$x \mapsto ax + b$, $a > 0$	°C and °F differ in zero point and unit
	ratio	$x \mapsto ax$, $a > 0$	length in metres or in feet

INTUITION

A statement is **meaningful** on a scale if no permissible transformation can change its truth. "Twice as long" survives metres → feet; "twice as warm" does not survive °C → °F.

Speaker notes

Each scale is characterised by the transformations that preserve its information: any one-to-one relabelling for nominal, any strictly increasing function for ordinal, positive affine maps $x \mapsto ax + b$ ($a > 0$) for interval (°C to °F), and positive scalings $x \mapsto ax$ for ratio (metres to feet). The smaller the set of permissible transformations, the more structure the values carry, and the more statements are meaningful.

The criterion explains the statistics column of the previous table. The mean of an ordinal attribute is not meaningful because a strictly increasing recoding (1, 2, 3 to 1, 5, 9) can change which of two groups has the larger mean; the median's position is preserved by every such recoding. A ratio of two temperatures in °C changes under the affine map to °F (20 °C / 10 °C = 2, but 68 °F / 50 °F = 1.36); a ratio of two lengths is the same in metres and feet.

Why the Type Decides the Method

- ✗ **The classic error:** averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.
- ✗ **The subtle one:** 20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

> The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every method in this course depends on it.

Mini-Exercise

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

Speaker notes

Satisfaction in stars is ordinal. The mean treats the gap from one to two stars as equal to the gap from four to five, which the scale does not guarantee; the median and the mode use only the order and the labels. Means of rating scales are nonetheless widely reported in practice, which amounts to treating the scale as interval.

Year of birth is interval: year 0 is a convention, so "born twice as late" is meaningless, but the mean year and differences between years are meaningful. A difference of two interval values, such as an age computed from years of birth, is on a ratio scale. Payment method is nominal (only counts, shares and the mode are meaningful); income is ratio, with a true zero, so "twice the income" is a valid statement.

Mini-Exercise

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

Satisfaction is **ordinal**: its mean assumes equal gaps between the stars — report the median. Year of birth is **interval**: the mean year is fine, but ratios are not ("born twice as late" means nothing); differences such as age are fine.

Takeaway

> An attribute's **scale of measurement** licenses the operations you may perform on it. Choosing a statistic before checking the scale is the most common silent error in data analysis.

A Table Is a Sample From a Joint Distribution

customer	payment	satisfaction	born	income (k€)	returned?
1	card	4	1987	52	yes
2	invoice	2	1995	31	no
3	card	5	1979	78	no
…	…	…	…	…	…

Each column is a random variable

$X_{\text{payment}}, X_{\text{satisf.}}, \ldots$ — a function from customers to values, exactly as a random variable was defined.

Each row is one draw

The whole table is n draws from one joint distribution $p(x_1, \ldots, x_d)$ that nobody has written down.

Speaker notes

A column is a random variable in the formal sense: a function from the population (here, customers) to values. Drawing a row means drawing a member of the population and recording all its attribute values at once, so the row is one realisation of the vector $(X_1, \ldots, X_d)$. The customer number is an identifier, not an attribute to be modelled.

A dataset is not the distribution but a sample from it. Every quantity computed from the table — a mean, a proportion, a correlation — is therefore an estimate of a property of $p(x_1, \ldots, x_d)$. A different sample from the same distribution would give a different value, and this sample-to-sample variation is the estimate's error.

The Scale Suggests the Model

attribute	scale	a first model
returned?	nominal, two values	**Bernoulli**(p) — one parameter
payment	nominal, k values	**categorical**$(p_1, \ldots, p_k)$, $\sum_j p_j = 1$
satisfaction	ordinal	categorical — the order is not used, or an ordinal model
born	interval	normal, on a restricted range
income	ratio, positive, skewed	log-normal, or normal on $\log$ income

> The scale limits the model as it limits the statistics: a normal distribution for ZIP codes is as meaningless as their mean.

Speaker notes

The categorical distribution assigns $P(X = j) = p_j$ to each of k values, with $k - 1$ free parameters because $\sum_j p_j = 1$; the Bernoulli is the case $k = 2$. Naive Bayes classifiers, for example, commonly model each nominal attribute this way, with one categorical distribution per class.

Ordinal attributes are the hard case. A categorical model ignores the order; a numeric model (e.g. normal) assumes equal gaps between levels, which the ordinal scale does not justify. Dedicated ordinal models, such as cumulative-logit models, use the order without assuming equal gaps. For year of birth a normal model is only an approximation, since its support is the whole real line. Income is positive and right-skewed, so a normal model would put probability on negative incomes; a log-normal model means that $\log$ income is normal. ZIP codes are nominal despite consisting of digits.

What "i.i.d." Claims

The rows $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **independent and identically distributed** (i.i.d.) if

independent	$p(\mathbf{x}_1, \ldots, \mathbf{x}_n) = \prod_i p_i(\mathbf{x}_i)$	*one customer's row tells nothing about another's*
identically distributed	$p_i = p$ for every row i	*every row comes from the **same** joint distribution*

Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

Speaker notes

Independence is a statement about rows, not columns: the attributes within a row may be strongly dependent (income and age, say). Under i.i.d. the probability of the whole dataset factorises, $p(\mathcal{D} \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$, which is the form every likelihood-based fitting method starts from.

Answer to the question. Daily sales of one shop are autocorrelated (one day predicts the next) and follow trends and seasons, so they are neither independent nor identically distributed. Two visits by the same customer are dependent because they share the person. Neighbouring pixels of one image are strongly correlated. A survey sent to a random sample yields approximately independent draws from the population (exactly so for sampling with replacement; without replacement the dependence is negligible when the population is large), though non-response can still make it unrepresentative. Record data is the setting where i.i.d. is at least plausible; graphs, grids and time series are the settings where it is not. A frequent violation of "identically distributed" in practice is distribution shift between the data a model is fitted on and the data it is applied to.

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Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

Only the random survey is plausibly i.i.d. Time series, repeated measurements and spatial data all violate it — the arrangement *is* information.

Takeaway

A dataset is n **draws from an unknown joint distribution**: columns are random variables, rows are draws, and the scale of each column suggests its model. Reading a table this way rests on an assumption — **i.i.d.** — that should be checked, not presumed.

Four Shapes of Data

Record data

Objects × attributes — the familiar table. Rows are *treated as* independent.

Graph data

Objects carry **links** to other objects: web pages, social networks, citations.

Fields / grids

Objects sit in a **spatial** arrangement: images, sensor lattices, simulations.

Spatio-temporal

Position **and** time: monthly land temperatures, traffic movement, trajectories.

> The shape decides which methods even apply. A table whose rows are treated as independent can be shuffled without loss; a grid or a time series cannot — the arrangement *is* information. Independence of rows is an assumption, not a property of the file.

Speaker notes

Ask what is lost by flattening each non-record type into a table. That question is the bridge to network science and to time-series methods, and it recurs whenever someone loads a CSV.

Three Data Quality Problems

Noise

Random distortion of measured values. Cannot be removed, only modelled or smoothed.

Outliers

Values far from the rest. Either an **error** to remove or the **signal** you were looking for — fraud, faults, discoveries.

Missing values

Absent entries. Handling depends on **why** they are missing, which the data itself does not tell you.

> "Outlier" is not a property of a data point; it is a judgement about a model of what is normal. Deleting outliers deletes evidence against that model.

Mini-Exercise

A weather sensor misses 8% of its temperature readings, and it fails mostly **when it is very cold**. Is deleting the rows with missing values harmless?

Speaker notes

This is Principle 3 in its most practical form: a preprocessing decision is a modelling assumption, and here an untestable one. Students should leave able to ask "why is this missing?" before "how do I impute it?". If time allows, ask what else could have told us about the mechanism (a thermometer next to it, the sensor's data sheet) — domain knowledge, not the data.

Mini-Exercise

A weather sensor misses 8% of its temperature readings, and it fails mostly **when it is very cold**. Is deleting the rows with missing values harmless?

> No. Deletion removes the coldest days, so every statistic afterwards — the mean temperature first — is biased upwards. Had the gaps been random transmission dropouts, deletion would be roughly harmless. The file looks the same in both cases; only the **mechanism** differs.

Mini-Exercise

A weather sensor misses 8% of its temperature readings, and it fails mostly **when it is very cold**. Is deleting the rows with missing values harmless?

No. Deletion removes the coldest days, so every statistic afterwards — the mean temperature first — is biased upwards. Had the gaps been random transmission dropouts, deletion would be roughly harmless. The file looks the same in both cases; only the **mechanism** differs.

The shape of the data decides which methods apply; the **reason** for a quality problem decides how it may be handled. Neither is visible in the file.

The Churn Shop's Customer Table

customer	contract	satisfaction	tenure (months)	postcode	churned?
1	monthly	4	3	94032	yes
2	yearly	—	27	80331	yes
3	yearly	5	41	94032	no
…	…	…	…	…	…

Speaker notes

The running example as a table: the same subscription business whose app users churn less (Lecture 01), whose gap was tested (Lecture 02) and predicted (Lectures 04 and 06). Every earlier lecture used this table without asking what its columns are.

The missing ratings are the lecture's point in one number. Dropping the rows with a missing rating leaves 60 churners among 915 customers (60/915 ≈ 6.6%): the missingness depends on the outcome itself, so the cleaned table is no longer a sample of the customers. Ask what a plot would have shown first — the share of missing ratings split by churned yes/no.

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…	…	…	…	…	…

SCALES

Contract, postcode, churned: nominal. Satisfaction: ordinal. Tenure: ratio. The mean postcode computes — and means nothing.

WHY IS SATISFACTION MISSING?

Customers who left stop answering: 40 of 100 churners gave no rating, 45 of 900 stayers. Drop those rows, and churn falls from **10%** to **6.6%**.



Module 2

Seeing the Data

> ***Guiding Question:*** *Summary statistics answer the questions you thought to ask. What shows you the ones you did not?*

Visualization Is Everywhere

Newspaper tables, subway maps, weather charts, stock graphs, a 3-D reconstruction of an injured knee, demographic bar charts, social-network layouts.

What do these have in common? All of them present data in a form the eye processes faster than the equivalent numbers.

Speaker notes

Question-first (P1). Let students name the common property before it is stated; "you can see it at a glance" is the answer, and the next slide gives the reason.

Why It Works

Bandwidth

Vision is the human sense with the highest data rate — pattern detection runs in parallel and pre-attentively.

Interaction

A picture is a natural surface for exploring: zoom, filter, select, compare.

Discovery

Visualization surfaces the **unexpected** — the structure nobody thought to compute a statistic for.

> Summary statistics answer questions you already asked. A plot lets the data ask you one — which is exactly what Anscombe's quartet demonstrated.

> Visualization is not decoration for results; it is an **instrument of analysis**, and the only one that can show you what you did not think to measure.

Speaker notes

Deliberate callback to L01 and L02 (spiral): the same argument, now with the tools to act on it. This unit is the hinge between "data does not speak for itself" and the concrete plot types that follow.

The Boxplot

A **visual summary of a distribution built from quartiles**:

The box

Spans Q_1 to Q_3 — the interquartile range, the middle half of the data.

The line

The **median** inside the box. Its position shows skew.

The whiskers

(Tukey) Reach to the most extreme data points inside $[Q_1 - 1.5 \cdot \mathrm{IQR}, Q_3 + 1.5 \cdot \mathrm{IQR}]$. Points outside are drawn singly as **outliers**.

The 1.5 × IQR rule is a **convention**, not a test. It flags candidates for inspection, not errors for deletion.

Speaker notes

Boxplots are built entirely from the quantiles the students defined in Lecture 02 — say so. Their strength is comparison: ten boxplots side by side are readable where ten histograms are not.

The Histogram

What it shows

The **shape** of a distribution: modality, skew, gaps and clusters that a boxplot hides entirely.

What it hides

Everything depends on the **bin width**. Too wide and structure vanishes; too narrow and noise looks like structure.

A bimodal distribution and a unimodal one can produce the same boxplot. Always look at one histogram before trusting a five-number summary.

Mini-Exercise

Two datasets have the same minimum, Q_1, median, Q_3 and maximum — so the same boxplot. Must their histograms look alike?

Speaker notes

If there is a minute left, let one student sketch the two-cluster version on the board. Some tools draw the whiskers to the minimum and maximum instead of the Tukey fences; always check which convention a plot uses.

Mini-Exercise

Two datasets have the same minimum, Q_1, median, Q_3 and maximum — so the same boxplot. Must their histograms look alike?

> No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

Mini-Exercise

Two datasets have the same minimum, Q_1, median, Q_3 and maximum — so the same boxplot. Must their histograms look alike?

No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

The Scatterplot

An X–Y plot of two dimensions against each other — the standard instrument for inspecting a **relationship** rather than a distribution.

With more than two dimensions there are four options:

Subsetting

Let the user choose which dimensions to display.

Reduction

Project to fewer dimensions (see preprocessing, and projections later).

Embedding

Map extra dimensions to other channels: colour, size, shape.

Multiple displays

Superimpose, or juxtapose in a **scatterplot matrix**.

Speaker notes

The four options are exhaustive and worth memorising — every high-dimensional visualization technique is one of them or a combination. Colour and size are channels with far lower precision than position; use them for secondary variables.

What a Scatterplot Reveals

Linear association, non-linear structure, clusters, outliers and heteroscedasticity — most of which no single correlation coefficient reports.

Pearson correlation $r = \dfrac{\operatorname{cov}(X, Y)}{s_X s_Y} \in [-1, 1]$: the covariance measured in units of the two standard deviations. It measures **linear** association only.

A correlation of 0 is consistent with a perfect parabola. The scatterplot shows it; the coefficient does not.

Tufte's Principle

Maximise the data-ink ratio: every drop of ink on the page should carry information. Delete anything that does not — heavy gridlines, boxes, redundant tick marks, decorative fills. **Range frames** apply the principle to the axes: draw each axis only over the interval the data actually spans, so the frame itself reports the minimum and maximum.

> The exercise: take a finished chart and ask of each element, *what would I lose by deleting this?*

A scatterplot shows the structure a coefficient summarises away. Beyond two dimensions you must **subset, reduce, embed or juxtapose** — and every mark you draw should earn its ink.

Speaker notes

Tufte's rule is a good design heuristic and an over-applied dogma; note that some redundancy aids reading under time pressure. The point students should keep is intentionality: each element defended, not each element removed.

When a Line Is Justified

Connecting points with a line asserts an **order** between them — and often an interpolation.

> **Data requirement:** one or more quantitative value attributes on a comparable scale, plus one **ordered** key attribute (usually time).

> Connecting nominal categories with a line claims an order that does not exist. The line is a statement, not a decoration.

Speaker notes

This is the attribute-types unit paying off immediately: the permissible plot follows from the scale of the key attribute. Bar charts for nominal, lines for ordered.

Aspect Ratio Is Not Cosmetic

Readers compare trends by the **orientation** of the line segments, so the shape of the plotting area changes what they conclude. The same series looks flat in a wide frame and dramatic in a tall one.

Banking to 45° chooses the aspect ratio so that the segments' average absolute orientation is 45°: differences in orientation are judged best near ±45° (Cleveland et al. 1988).

Additional channels

Colour and line type distinguish series — but only for a handful of them.

Multiple Series

Overlaid

Best for comparing a few series directly against each other.

Stacked

Each series is drawn on top of the previous one. Shows the total well — and makes individual series **hard to read**, since each sits on a moving baseline.

Sorted / small multiples

One panel per series, ordered by some property. Scales to many series where overlaying fails.

A line asserts order and interpolation; the **aspect ratio** decides what the reader sees and stacking trades the readability of the parts for the readability of the whole.

Speaker notes

The stacked-plot critique is worth demonstrating: ask students to read the third band's trend off a stacked chart — they cannot, because the baseline moves. Small multiples are the usual better answer.

Module 3

Preprocessing

> ***Guiding Question:*** *Lecture 01 claimed that preprocessing is modelling. Which choices made here change what a model can learn?*

Aggregation

Combine several objects or attributes into one — customer purchases summarised by store, by month, by price band.

Gains

Fewer objects to process; **less variation**, hence more stable estimates; the natural operation of a data warehouse.

Costs

Detail is gone, irreversibly. Effects that exist only within a group become invisible.

Speaker notes

Aggregation is where Simpson's paradox lives: a trend can reverse when groups are pooled. Worth one sentence — students will meet it in dashboards long before they meet it in a statistics course.

Sampling

Work with a subset of the objects: random sampling with or without replacement, or **stratified** sampling that preserves group proportions.
Sampling always loses information. The question is whether it loses anything that matters — a sufficiently large sample retains the overall structure.

> Sample a sine wave once per period and every sample has the same value: the reconstruction is a flat line — not merely rougher, but **wrong**.

Speaker notes

The sine example is aliasing: too sparse a sample does not blur a pattern, it produces a different one. The same holds for customer data sampled once a month when the behaviour of interest is weekly.

Scaling and Encoding

Scaling numeric attributes

Min–max: $x' = \dfrac{x - \min}{\max - \min} \in [0, 1]$

Standardisation: $z = \dfrac{x - \bar{x}}{s}$ — mean 0, standard deviation 1

Encoding categorical attributes

Nominal → one-hot: one 0/1 column per value, so no false order.

Ordinal → ranks $1, 2, 3, \ldots$, which keep the order.

Two customers differ by 1,000 € in income and 30 years in age. Unscaled, their distance is $\approx 1{,}000$: age is invisible. Standardised ($s = 10{,}000$ € and 15 years) the differences are 0.1 and 2 — now age dominates.

Speaker notes

Scaling is a modelling decision in the sense of this lecture: it decides which attribute a distance listens to. It matters for every distance-based method (clustering, nearest neighbours) and for PCA, which would otherwise find "the direction of income". Encoding nominal values as 1, 2, 3 invents an order and a distance between categories; one-hot avoids it at the cost of more columns. Exact numbers: $\sqrt{(1000^2 + 30^2)} \approx 1000.4$ and $\sqrt{(0.1^2 + 2^2)} \approx 2.0$.

Dimensionality Reduction

The curse of dimensionality. As the number of attributes grows, the data becomes sparse: distances between points concentrate, neighbourhoods lose meaning, and the volume to be covered grows exponentially.

The problem

Many features are irrelevant, noisy or correlated.

The cost

More computation, more parameters, more variance — and weaker distance-based methods.

The response

Feature selection keeps a subset; **feature extraction** builds new combined dimensions.

Every preprocessing step **trades information for tractability**. The trade is usually worth it — but it is a modelling decision, and it should be recorded as one.

Speaker notes

Point forward: projections (PCA, MDS) are the formal treatment of feature extraction, and clustering suffers directly from the curse — and from unscaled attributes. This unit is the motivation both later lectures rely on.

Image Credits & References

Cleveland, William S. and McGill, Marylyn E. and McGill, Robert (1988). The Shape Parameter of a Two-Variable Graph. *Journal of the American Statistical Association*.